

CSA1717-ARTIFICIAL INTELLIGENCE
PRACTICAL LAB EXPERIMENTS
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Aim:

To write a Prolog program to find the **number of vowels in a given string**.

Objective:

To identify vowels (a, e, i, o, u) from a string and count their total number using recursion.

Algorithm:

1. ☐ Start the program.
2. ☐ Read a string as a list of characters.
3. ☐ Take the first character.
4. ☐ Check whether the character is a vowel.
5. ☐ If it is a vowel, increment the count.
6. ☐ If it is not a vowel, continue without incrementing.
7. ☐ Recursively process the remaining characters.
8. ☐ When the list becomes empty, return the total count.
9. ☐ Stop.

Prolog Program:

% Program to count vowels

vowel(a).

vowel(e).

vowel(i).

vowel(o).

vowel(u).

count_vowels([], 0).

count_vowels([H|T], Count) :-

 vowel(H),

 count_vowels(T, Count1),

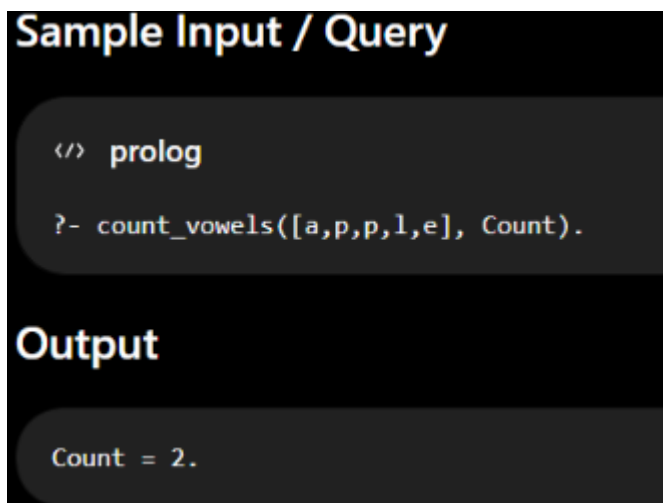
 Count is Count1 + 1.

count_vowels([H|T], Count) :-

 \+ vowel(H),

 count_vowels(T, Count)..

Output

A screenshot of a Prolog interpreter interface. It features a dark background with light-colored text. The top section is titled "Sample Input / Query" in a bold, light-colored font. Below this title, there is a light gray rounded rectangle containing two lines of Prolog code: a prolog declaration and a query. The bottom section is titled "Output" in a bold, light-colored font. Below this title, there is a light gray rounded rectangle containing the output result.

```
</> prolog
?- count_vowels([a,p,p,l,e], Count).
```

Output

```
Count = 2.
```

Result:

Thus, the Monkey Banana Problem is successfully represented using Prolog.